# LittleSteps — Learning & Play App for Kids

Master task brief · Product spec v1.2 · Platforms: iOS & Android (tablet first)

## 1. Overview

LittleSteps is a mobile app for children aged 4–8 that teaches early reading, counting and shapes through short mini-games (2–4 minutes each). Children earn stickers for finishing activities and collect them in a sticker book. Parents get a separate, protected area to follow progress, set daily screen-time limits and choose which subjects are available.

## 2. Goals

- Launch with 3 subjects (Reading, Numbers, Shapes) and at least 30 mini-games.
- A child can start playing within 2 taps from opening the app, without reading text.
- Full offline play for downloaded subjects; progress syncs when back online.
- Compliance with COPPA and GDPR-K: no ads, no third-party trackers, no chat.

## 3. Target users

- **Child (4–8):** may not read yet; needs large touch targets (min 64 px), audio narration and icons.
- **Parent:** creates the account, manages up to 4 child profiles, reviews weekly progress.
- **Teacher (later phase):** classroom mode, out of scope for v1.

## 4. In scope / out of scope

| In scope (v1) | Out of scope (v1) |
|---|---|
| Child profiles with avatar selection | Classroom / teacher accounts |
| 30 mini-games across 3 subjects, audio narration in English and Turkish | Multiplayer or chat of any kind |
| Sticker rewards and sticker book | In-app purchases inside child mode |
| Parent gate, parent dashboard, screen-time limits | Web version |
| Offline mode with background sync | Smart-watch companion app |

## 5. Success metrics

- Day-7 retention of child profiles ≥ 35%.
- Average session length between 8 and 15 minutes (screen-time limits respected).
- Crash-free sessions ≥ 99.5%; app start to first game < 3 seconds on mid-range tablets.

## 6. Functional requirements

- **Profiles:** parent creates child profiles (name, age band, avatar). No photos of children are stored.
- **Mini-games:** each game has 5 rounds, adapts difficulty after 2 correct or 2 wrong answers in a row, and gives spoken feedback. Games are data-driven (JSON definitions + assets) so new games ship without an app update.
- **Rewards:** one sticker per finished game, a special sticker every 5 games; stickers can be placed in a sticker book.
- **Parent gate:** entering the parent area needs an adult challenge (e.g. “press and hold the three blue squares”) plus the parent PIN.
- **Screen time:** parent sets a daily limit (15–90 min); at the limit a friendly “time to rest” screen appears and games lock until the next day.
- **Progress:** per-skill mastery (0–100%) and a weekly summary for parents, optionally by e-mail.

The user flow below was sketched in the kickoff workshop (photo of the whiteboard).

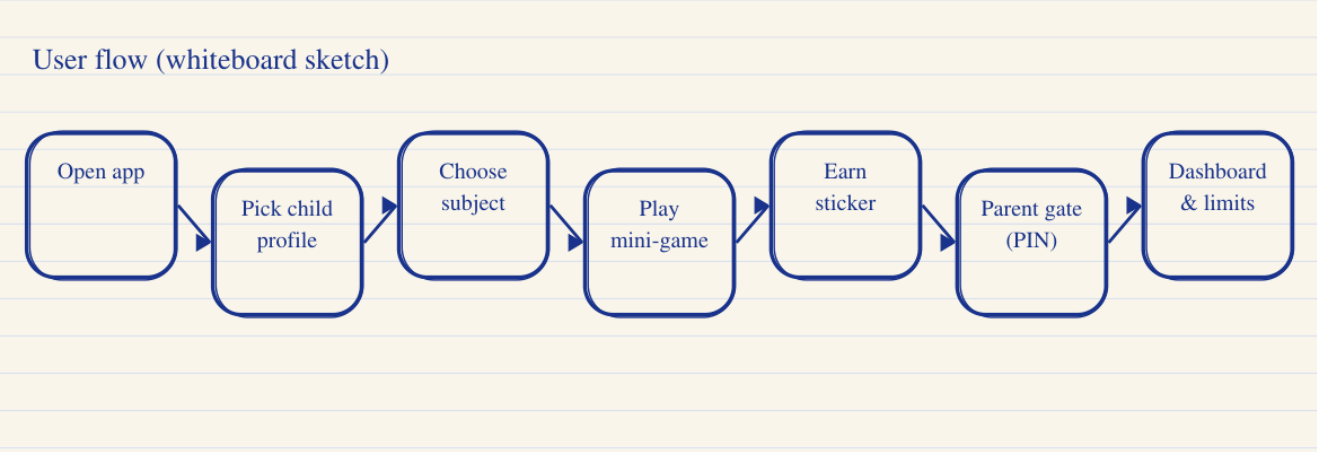



Figure 1. Child and parent user flow (whiteboard photo from kickoff workshop).

## 7. Non-functional requirements

- Accessibility: all instructions are spoken; colour is never the only signal; supports switch control.
- Privacy: data minimisation, all child data encrypted at rest; analytics are anonymous and first-party only.
- Performance: game assets ≤ 15 MB per subject pack, downloaded on Wi-Fi by default.
- Content safety: every new game passes an educational review and a child-safety review before release.

## 8. Key screens (wireframes)

Low-fidelity wireframes for the three most important screens. Large buttons, minimal text, every button also has an icon and a spoken label.

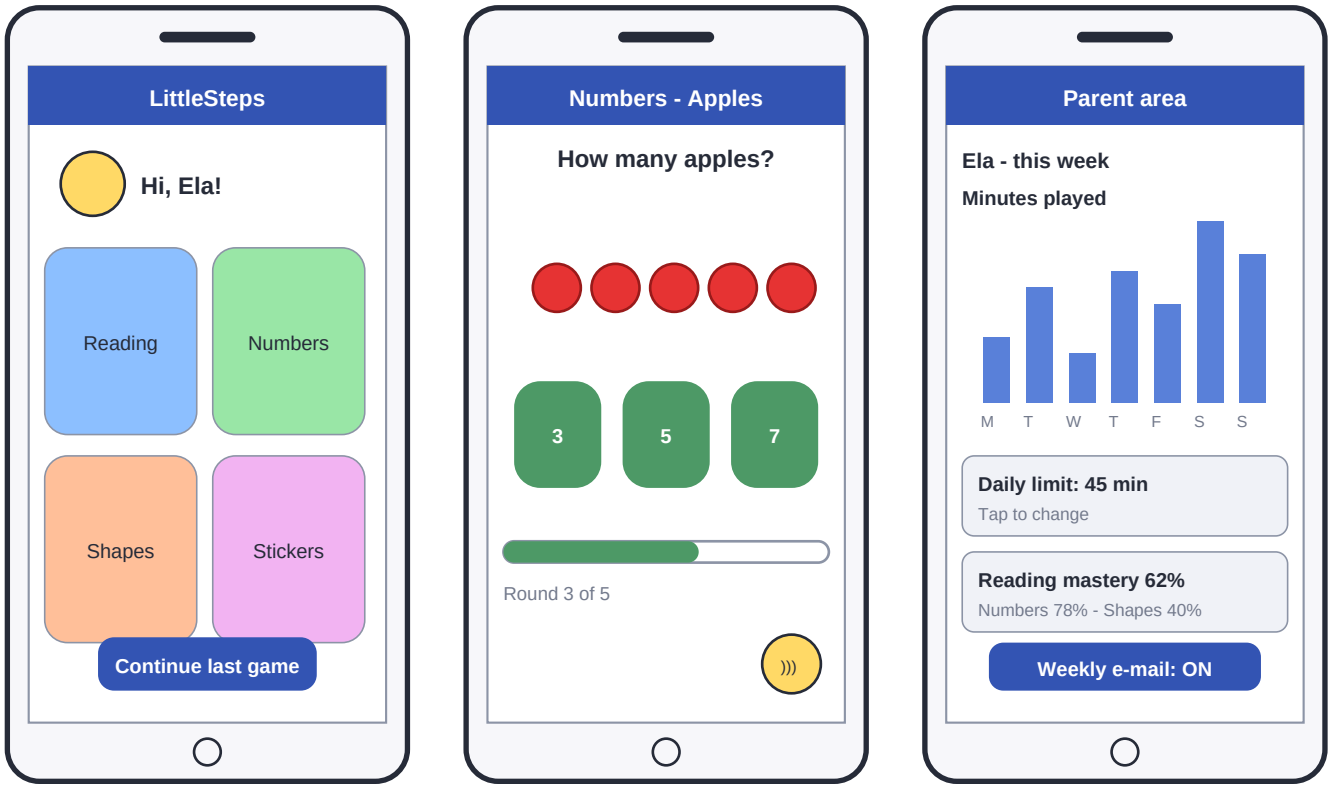



Figure 2. Wireframes: child home screen, counting mini-game, parent dashboard.

## 9. Content plan

| Subject | Games at launch | Skills |
|---|---|---|
| Reading | 12 | Letter sounds, first words, rhyming, picture-word matching |
| Numbers | 10 | Counting to 20, more/less, simple addition with objects |
| Shapes | 8 | 2D shapes, colours, patterns, simple puzzles |

Narration is recorded by voice actors in English and Turkish. Every game definition contains the audio keys so translations can be added later without code changes.

## 10. System architecture

Flutter app with a local SQLite store for offline play. A small backend handles accounts, progress sync and content packs. No third-party SDKs that collect data.

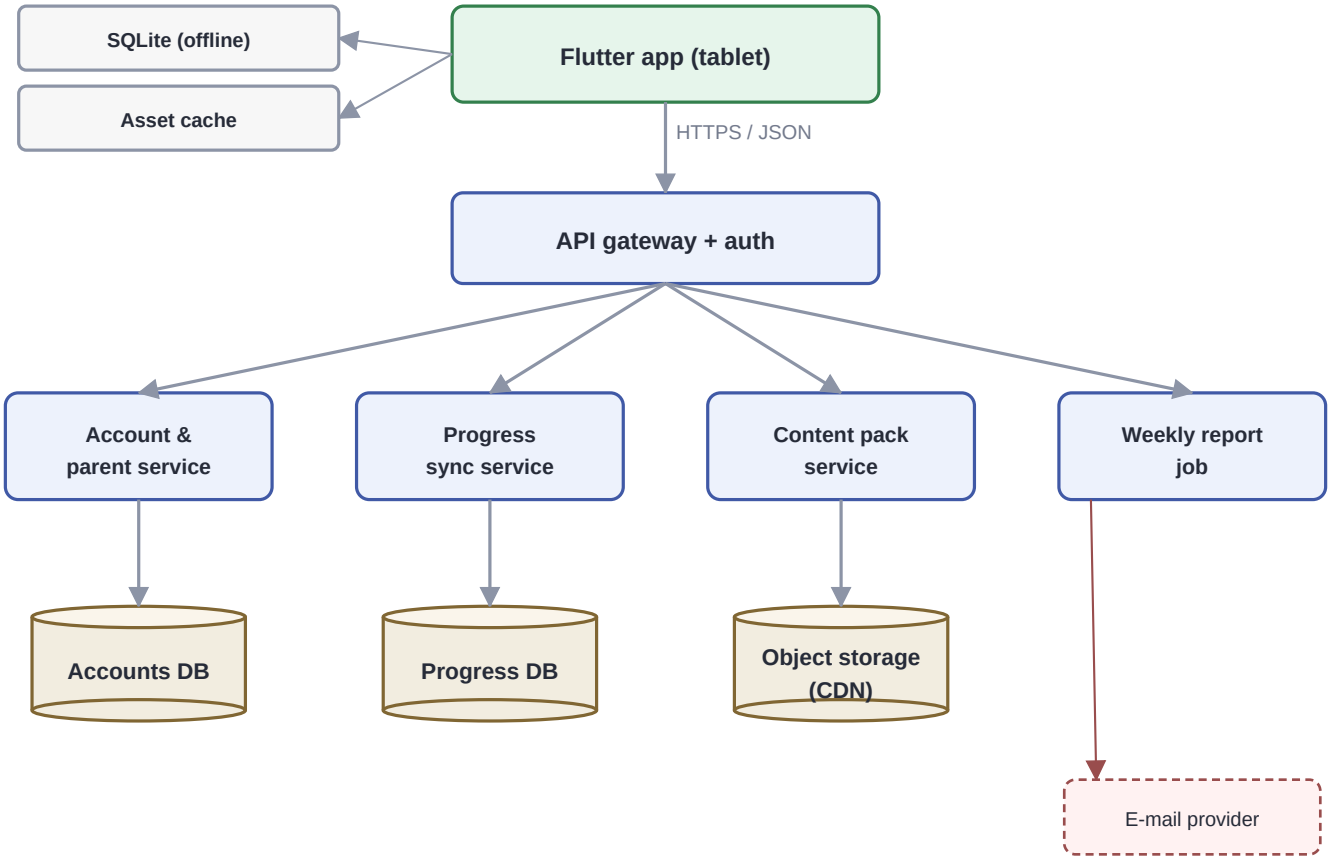
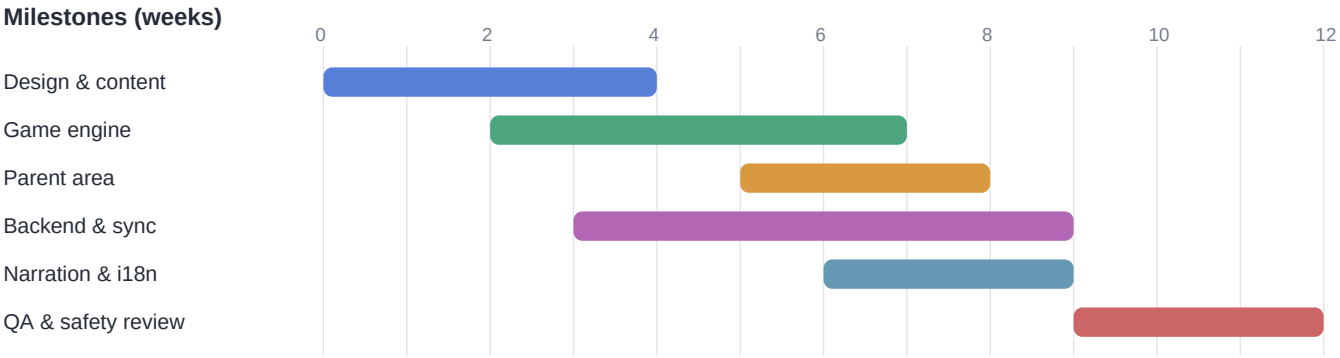



Figure 3. System architecture.




## 11. Open questions

- Should parents be able to buy extra subject packs, and where (outside child mode only)?
- Is e-mail the right channel for the weekly report, or push notifications to the parent device?